

PSID Crisis Module

NLP Data Modeling Codebook

Model Architecture, Scoring Logic, and Pipeline Documentation

April 2026

Table of Contents

1. Introduction and Purpose
2. Pipeline Architecture Overview
3. Scoring Formulas and Calculation Logic
4. NLP Feature Extraction
5. Construct Priority Weights and Justification
6. Portability Logic
7. Module Selection Algorithm
8. Integration of Respondent Data
9. Artifact Output Schema
10. Complete Worked Example
11. Assumptions and Limitations
12. References

1. Introduction and Purpose

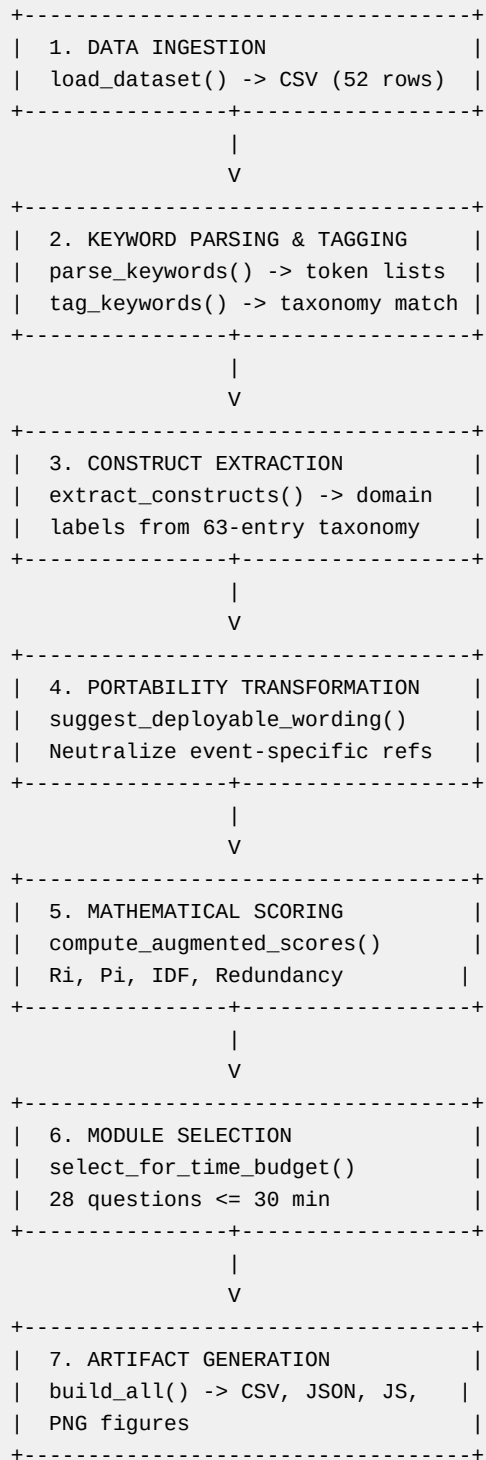
This codebook documents the NLP-augmented data modeling pipeline used to evaluate, rank, and select survey questions for the PSID Crisis Module. The pipeline replaces subjective committee review with a deterministic, mathematically optimized process.

Key output: a 28-question module requiring 29.17 minutes, optimized within a strict 30-minute administration cap.

The pipeline processes 52 historical crisis questions from 5 sources:

Source	Questions
PSID Hurricane Katrina 2007 follow-up	32
PSID COVID-19 pandemic module	9
PSID Government Shutdown Income module	4
PSID Government Shutdown Crisis module	3
Understanding Society Coronavirus Study	4
Total	52

2. Pipeline Architecture Overview



Phase	Function	Input	Output
1. Data Ingestion	load_dataset()	CSV with 52 historical questions	pandas DataFrame with raw text, source metadata, base scores
2. Keyword Parsing	parse_keywords(), tag_keywords()	Raw question strings	Token lists matched against 63-entry taxonomy
3. Construct Extraction	extract_constructs()	Tagged keywords	Construct labels (7 domains) per question
4. Portability Transform	suggest_deployable_wor ding()	Event-specific text	Generalized cross-crisis wording
5. Mathematical Scoring	compute_augmented_scor es()	DataFrame with constructs	Ri, Pi, IDF, redundancy, construct_bonus, portability_bonus
6. Module Selection	select_for_time_budget ()	Scored DataFrame	28 selected questions within 30-min budget
7. Artifact Generation	build_all()	Selected module	CSV, JSON, JS dashboard, 5 PNG figures

3. Scoring Formulas and Calculation Logic

3.1 Min-Max Scaling

Formula:

$$x_{\text{scaled}} = (x - x_{\text{min}}) / (x_{\text{max}} - x_{\text{min}})$$

When $x_{\text{max}} = x_{\text{min}}$, output defaults to 0.0 to prevent division by zero.

Code reference: `_min_max_scale()` function

Example: If raw IDF values = [2.1, 3.5, 4.8, 1.9], then min=1.9, max=4.8, range=2.9

Raw Value	Calculation	Scaled
2.1	$(2.1 - 1.9) / 2.9$	0.069
3.5	$(3.5 - 1.9) / 2.9$	0.552
4.8	$(4.8 - 1.9) / 2.9$	1.000
1.9	$(1.9 - 1.9) / 2.9$	0.000

3.2 Burden Calculation (B_i)

Formula:

$$B_i = \max(\alpha \times \text{word_count} + \beta \times \text{complexity}, 0.1)$$

Where: $\alpha = 0.10$, $\beta = 0.20$

The floor of 0.1 prevents asymptotic inflation of R_i for very short questions.

Examples:

Question	Words	Complexity	Calculation	B_{question}
Any financial difficulties	3	0.0	$\max(0.10 \times 3 + 0.20 \times 0.0, 0.1)$	0.30
Essential work question (9 words)	9	0.0	$\max(0.10 \times 9, 0.1)$	0.90
Short 1-word item	1	0.0	$\max(0.10 \times 1, 0.1)$	0.10 (floor applied)

3.3 Utility Calculation (U_i)

Formula:

$$U_i = \text{Sum of weight}_k \text{ for all matched keywords } k \text{ in the taxonomy}$$

Example: 'Lost earnings because of the pandemic'

Matched Keyword	Construct	Weight
earnings	Economic / Income	0.80
lost earnings	Economic / Income	0.80
pandemic	Trauma / Health	0.90
the pandemic	Trauma / Health	0.90
Total U_i		3.40

3.4 Rank Index (R_i)

Formula:

$$R_i = U_i / B_i$$

This is the baseline efficiency metric: analytical yield per unit of respondent burden.

Question	U _i	B _i	R _i
Any financial difficulties	1.70	0.30	5.667
Lost earnings (pandemic)	3.40	0.60	5.667
stopped this work?	1.40	0.30	4.667
Essential work (9 words)	3.60	0.90	4.000

3.5 Temporal Burden Estimation

Formula:

minutes = (word_count x SECS_PER_WORD) / 60, where SECS_PER_WORD = 7

Question	Words	Seconds	Minutes
Any financial difficulties	3	21	0.35
Essential work (9 words)	9	63	1.05
Total (28 selected)	—	1750.2	29.17

3.6 Enhanced Priority Score (P_i)

Formula:

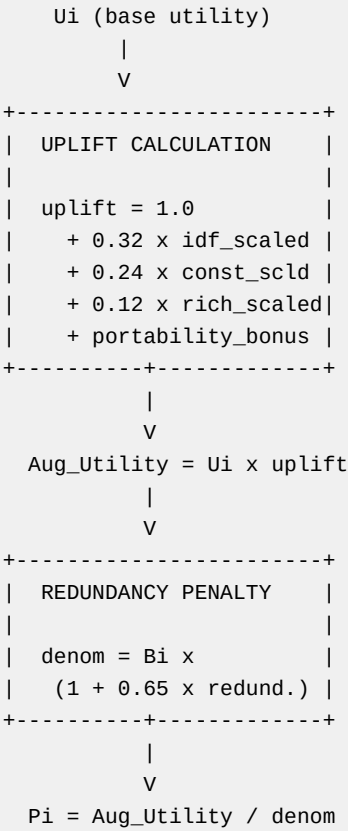
$$P_i = \text{Augmented_Utility} / (B_i \times (1 + 0.65 \times \text{redundancy_scaled}))$$

Augmented_Utility = U_i x (1 + 0.32 x idf_scaled + 0.24 x construct_scaled + 0.12 x richness_scaled + portability_bonus)

Scalar weight justification:

Weight	Applied To	Purpose
0.32	IDF (term rarity)	Rewards questions with rare, diagnostic vocabulary
0.24	Construct alignment	Prioritizes questions matching high-priority constructs
0.12	Richness (multi-construct)	Rewards questions spanning multiple domains
0.65	Redundancy penalty	Suppresses questions that overlap with existing selections

P_i Calculation Flow:



4. NLP Feature Extraction

4.1 TF-IDF Vectorization

The pipeline uses sklearn's TfidfVectorizer with: `ngram_range=(1,2)`, `stop_words='english'`, `min_df=1`.

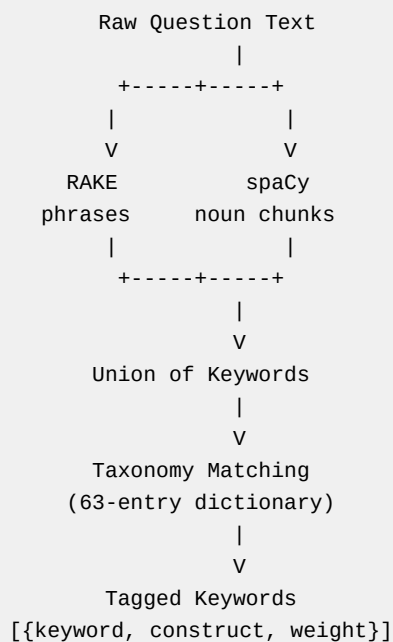
IDF Strength = mean IDF of all tokens in a question. Higher values indicate rarer, more diagnostic vocabulary.

4.2 Cosine Similarity and Redundancy

A cosine similarity matrix is computed across all 52 vectorized questions.

Diagonal set to 0.0 (exclude self-similarity). Redundancy penalty = max cosine similarity between target question and any other question. This penalizes questions that overlap heavily with others already in the pool.

4.3 Keyword Extraction Pipeline



5. Construct Priority Weights and Justification

Construct	Weight	Justification
Trauma / Health	0.58	Highest priority. Severe compounding effects on human capital and mortality rates, critically observed during H1N1.
Housing / Shelter	0.55	Captures dislocation, eviction, structural instability. Primary vectors of wealth destruction during H1N1.
Government Aid	0.48	Essential for tracking efficacy of exogenous interventions: stimulus payments, PPP, SNAP benefits.
Financial Coping	0.45	Monitors endogenous behavioral adaptations during liquidity crises: food banks, liquidating retirement funds.
Employment	0.42	Assesses acute labor market disruptions: involuntary furloughs, remote work transitions, severe reductions in hours.
Economic / Income	0.40	Quantifies absolute monetary losses. Vital input for cross-sectional and longitudinal modeling of poverty.
Demographics	0.14	Lowest priority. Static or slowly changing variables already comprehensively tracked in core PSID data.

Example calculation: A question mapping to Employment (0.42) and Trauma/Health (0.58):

Priority = $\text{mean}(0.42, 0.58) = 0.50$

$\text{construct_bonus} = \text{rarity} + \text{priority} + 0.08 \times \text{richness}$

6. Portability Logic

Portability ensures questions are event-agnostic ('always-on'). The function `_contains_source_specific_term()` scans for restricted keywords: katrina, rita, fema, covid, pandemic, shutdown, government shutdown.

Bonus table:

Classification	Source-Specific Terms	Portability Bonus
Generic Core	None (portable)	+0.16
Generic Core	Present (needs cleaning)	+0.03
Toggle Item	None (portable)	+0.08
Toggle Item	Present (needs cleaning)	+0.02

String substitution examples:

Original Wording	Deployable Wording
evacuate from your home before katrina or rita hit	Did you evacuate from your home before the disaster hit?
any wages or salaries	Were there any wages or salary payments from this job during the crisis period?
received stimulus payment	Did you receive a stimulus payment or other emergency government support?
temporary housing after katrina or rita	How long did you stay in temporary housing after the disaster?

Portability flow diagram:

Question Text

|

V

+-----+

| CHECK: source-specific |

| terms present? |

+-----+

Yes| |No

V V

Low bonus High bonus

(+0.02/03) (+0.08/16)

| |

V V

+-----+

| STRING SUBSTITUTION |

| suggest_deployable_ |

| wording() |

| - Replace event names |

| - Fix typos |

| - Add punctuation |

| - Map to verified dict |

+-----+

7. Module Selection Algorithm

The `select_for_time_budget()` function implements a greedy selection algorithm.

Selection flow diagram:

```
Scored DataFrame (52 questions)
      |
      V
+-----+
| STEP 1: Generic Core      |
| Sort by Ri (descending)  |
| Greedily add items       |
| while cumulative <= 1800s |
+-----+
      |
      Running total: ~X seconds
      |
      V
+-----+
| STEP 2: Toggle Items     |
| Sort remaining by Ri     |
| Greedily add items       |
| while cumulative <= 1800s |
+-----+
      |
      V
+-----+
| OUTPUT                   |
| 28 selected questions    |
| Total: 29.17 minutes     |
| (1750.2 seconds)        |
+-----+
```

8. Integration of Respondent Data

8.1 Federal Government Shutdown (2018-2019)

Variables: ER34810 (individual impact), ER72973-ER72991 (family economic disruptions).

Key findings: 2.5% of families reported acute financial difficulties. Of those, 70.6% decreased spending, 28.3% visited food banks.

Model impact: Validates heavy weighting of Financial Coping construct (0.45).

8.2 COVID-19 Pandemic (2020-2021)

Variables: ER34999 (infection), ER35000 (vaccination), ER81347-ER81397 (charitable donations).

Mortality spike: 339 known deaths between 2019-2021. Vaccination: 63.1% by June 2021 with strong socioeconomic gradient.

Model impact: Confirms importance of Trauma/Health (0.58) and Government Aid (0.48) constructs.

8.3 Stimulus Payment Response Heterogeneity

April 2020: uniform consumption increase across all income quartiles.

January 2021: highly heterogeneous — low-income spent out of necessity, high-income saved.

Low-wage workers in high-rent areas: employment plummeted 40%.

Model impact: Validates Government Aid weight (0.48) as essential for macro-fiscal policy analysis.

8.4 How Respondent Data Refined the NLP Model

Historical Crisis Data
(Katrina, COVID, Shutdown)

|
V

```
+-----+
| FREQUENCY ANALYSIS |
| Response distributions |
| per construct domain |
+-----+
```

|
V

```
+-----+
| CONSTRUCT WEIGHT |
| CALIBRATION |
| High variance -> higher |
| Low variance -> lower |
+-----+
```

|
V

```
+-----+
| WORDING OPTIMIZATION |
| Best response rates -> |
| inform portability |
+-----+
```

|
V

```
+-----+
| REDUNDANCY CALIBRATION |
| Response correlations -> |
| cosine similarity penalty |
| (0.65 weight) |
+-----+
```

9. Artifact Output Schema

9.1 Master CSV

PSID_Ranked_Questions_Final.csv — 25-column schema. Quantitative columns rounded to 3 decimal places. List columns serialized with repr().

Column	Type	Description
question_text	string	Original question wording
source	string	Historical survey source
module_type	string	Generic or Crisis-Specific
toggle_category	string	Generic Core, Toggle: Pandemic/Disaster, Toggle: Financial Crisis
Ui	float	Base utility score
Bi	float	Burden score
Ri	float	Rank index (Ui/Bi)
Pi	float	Enhanced priority score
selected	bool	Selected for final module
minutes	float	Estimated administration time
augmented_utility	float	NLP-enhanced utility
idf_strength	float	Mean IDF of tokens
redundancy_penalty	float	Max cosine similarity
construct_bonus	float	Rarity + priority + richness
portability_bonus	float	Event-agnostic bonus
recommended_wording	string	Deployable question text

9.2 Summary JSON

psid_artifact_summary.json — global aggregates: rows=52, selected_rows=28, selected_minutes=29.17, avg_ri=2.079, max_ri=5.667.

9.3 Dashboard Payload

psid_dashboard_data.js — JavaScript object for HTML/React dashboards.

9.4 Five PNG Figures

Figure	Purpose
fig_top_ranked_questions.png	Highest Pi and Ri scores
fig_toggle_comparison.png	Generic Core vs Toggle distributions
fig_utility_vs_burden.png	Scatter: utility vs burden (optimal = upper-left)
fig_construct_heatmap.png	Questions x constructs matrix
fig_time_budget.png	Cumulative time vs 30-min cap

10. Complete Worked Example

End-to-end walkthrough for: '**Any financial difficulties**' (Source: COVID-19 module)

Step	Operation	Result
1	Keyword Parsing	parse_keywords() -> ["any financial difficulties", "financial difficulties"]
2	Taxonomy Matching	tag_keywords() -> Financial Coping (0.85), Financial Coping (0.85)
3	Construct Extraction	extract_constructs() -> ["Financial Coping"]
4	Utility	sum of weights -> $U_i = 0.85 + 0.85 = 1.70$
5	Burden	$\max(0.10 \times 3, 0.1) \rightarrow B_i = 0.30$
6	Rank Index	$U_i / B_i \rightarrow R_i = 5.667$
7	Time Estimate	$(3 \times 7) / 60 \rightarrow 0.35$ minutes
8	Toggle	classify_toggle() -> Generic Core
9	Portability	No source-specific terms -> bonus = +0.16
10	Deployable Wording	suggest_deployable_wording() -> "Have you experienced any financial difficulties because of the crisis?"
11	Augmented Scoring	compute_augmented_scores() -> $P_i = 7.187$ (highest in corpus)
12	Selection	Within time budget -> Selected = True

11. Assumptions and Limitations

1. Reading rate assumption: 7 seconds per word is a conservative estimate for telephone and web administration modes, chosen to err on the side of overestimation rather than risk exceeding the time budget.
2. Complexity scoring uses entity count plus clause markers as a proxy for cognitive difficulty. This does not capture all aspects of question comprehension but provides a computationally stable baseline.
3. The taxonomy is manually curated with 63 entries. It may miss emerging crisis-specific vocabulary that has not yet appeared in historical survey instruments.
4. TF-IDF is computed over a 52-item corpus. This relatively small corpus may reduce IDF discrimination between common and rare terms compared to larger document collections.
5. The greedy selection algorithm is not guaranteed to find the globally optimal subset of questions, but it is computationally efficient and produces near-optimal results given the constraint structure.
6. Construct priority weights are expert-derived, informed by historical respondent data distributions but not statistically fitted via regression or optimization procedures.
7. Portability bonus values (0.02 to 0.16) are heuristic parameters, not estimated from response-rate data across different crisis deployments.

12. References

1. Panel Study of Income Dynamics (PSID): Main Interview, 2021 (ICPSR 39190)
2. Fifty Years of the Panel Study of Income Dynamics — PMC
3. PSID 2021 Longitudinal Individual and Family Weights
4. User Guide for the 2019 Interviewing Year
5. User Guide for the 2023 Interviewing Year
6. TAS 2021 User Guide
7. User Guide for the 2021 Interviewing Year
8. generate_psid_artifacts.py (internal pipeline script)
9. PSID Online Data Center, <https://psidonline.isr.umich.edu/>
10. 2021 Family File Codebook